Pulkit Katariya

+91-8868844188 | pulkit.kr1924@gmail.com | kryz73.github.io | github.com/kryz73 | linkedin.com/pulkit-katariya

Education

Indian Institute of Information Technology and Management, Gwalior Gwalior, India

B.Tech in Mathematics and Scientific Computing *CGPA: 8.11/10 | Expected: 2028*

- **Relevant Coursework:** Data Structures & Algorithms, Design and Analysis of Algorithms, Object-Oriented Programming, Operating Systems, Software Engineering, Database Systems, Computer Networks, Linear Algebra, Probability & Statistics
- **Certification:** WorldQuant University – Applied Data Science Lab

Technical Skills

Languages: C++, Go, Python, SQL, TypeScript, C, Bash
Backend & Architecture: REST APIs, Service-Oriented Architecture (SOA), Microservices, FastAPI, Strawberry GraphQL, Chi, Concurrency, JWT/RBAC
Databases & Storage: PostgreSQL, Redis (Caching), Apache Parquet, DuckDB, SQLite, Relational Modeling
Cloud & Reliability: Docker, Kubernetes, AWS (EC2, S3), GCP, CI/CD, GitHub Actions, Linux, systemd, Nginx, Git
Web & Frontend: React, Next.js, HTML5, CSS3, Tailwind CSS, Responsive Web Design
Applied ML & Analytics: XGBoost, PyTorch, Scikit-learn, Transformers (DistilBERT), TreeSHAP, MLflow

Open Source Contributions

Volcano – volcano-sh (CNCF Incubating Project)

Open Source Contributor | GitHub

- Authored comprehensive table-driven unit tests for all queue state transitions (`Open`, `Closed`, `Closing`, `Unknown`) in `pkg/controllers/queue/state/`, achieving 100% test coverage across the queue lifecycle for Volcano's HPC and AI batch scheduler.
- Engineered table-driven unit tests for all job controller state transitions in `pkg/controllers/job/state/`, achieving 100% statement coverage integrated into automated CI test suites to prevent regressions in distributed scheduling.

Projects

ACAD-ERP | Live Demo | GitHub | *FastAPI, Strawberry GraphQL, Next.js, React, PostgreSQL, Docker*

- Architected a modular enterprise platform across 9 domains, designing relational schemas in **PostgreSQL** enforcing strict relational constraints and transactional integrity (ACID).
- Implemented a production-grade authentication and authorization pipeline using **JWT HTTPBearer, Argon2 password hashing**, and live role hydration to secure role-gated workflows.
- Developed a unified **Strawberry GraphQL** API layer integrated with a responsive **Next.js / React** web client, eliminating over-fetching and ensuring deterministic payload contracts.

Harvest – YouTube Data Pipeline | GitHub | *Go, YouTube API, SQLite, Apache Parquet, AWS EC2, systemd*

- Architected an autonomous, fault-tolerant **Go ingestion daemon** deployed on **AWS EC2** with systemd supervision, structured logging, and automated recovery handling 24/7 continuous streaming data.
- Solved a strict **10,000 units/day** API rate limit across 3,500+ creator channels by engineering an async HTTP worker pool that prioritized due observations, achieving **0.02 quota units per snapshot**.
- Designed a 13-checkpoint non-uniform sampling state machine (0.5h–60h) with comment buffering, serializing sealed trajectories into **Snappy-compressed Apache Parquet** for zero-copy queries.

Viral – Predictive ML & Analytics | GitHub | *Python, XGBoost, DistilBERT, FastAPI, MLflow, TreeSHAP*

- Formulated a leakage-free temporal prediction pipeline analyzing **65,881 records and 4.1M+ comments**, evaluating early engagement kinematics and sentiment signals ($t \leq 6$h).
- Conducted 7 systematic ablation experiments; engineered dual-stage comment drift using **DistilBERT** (+21.3% PR-AUC lift) with an optimized **XGBoost** model achieving **0.6206 PR-AUC** and **0.9784 ROC-AUC**.
- Integrated **TreeSHAP** for feature attribution, tracked metrics in **MLflow**, and deployed an asynchronous **FastAPI** microservice delivering real-time predictions under a strict **sub-2ms SLA**.